

The empirical recurrence for OEIS A237103: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A237103 records “Empirical recurrence of order 54”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 147$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 54, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 54 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237103 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum minus the upper median of every 2×2 subblock equal.”. It has offset 1 and begins

1881, 39235, 848549, 20079181, 485177201, 11899510847, 293281801479, 7246691140947, ...

The entry states:

Empirical recurrence of order 54 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:51:21, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 147$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 147$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 294$ exact terms of the model returns order 54 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{54} c_i a(n-i),$$

c_1	63	c_{15}	−116698419004767	c_{29}	−3229888174215996700	c_{43}	1037673385778251264
c_2	−1265	c_{16}	−333295846044775	c_{30}	−15578117890377680352	c_{44}	209587940324148224
c_3	2536	c_{17}	1877110167281209	c_{31}	10600047341280264468	c_{45}	−188975696962109952
c_4	232630	c_{18}	865562189515044	c_{32}	19551994629745801472	c_{46}	−46846338597182976
c_5	−2562545	c_{19}	−17383061719272393	c_{33}	−18481230512994909480	c_{47}	20265582251368448
c_6	−6493583	c_{20}	12382958905852872	c_{34}	−17605836813814297880	c_{48}	6747364341858304
c_7	252951820	c_{21}	100760559174249059	c_{35}	21119879822315726096	c_{49}	−912786342008832
c_8	−833938136	c_{22}	−159177895459126802	c_{36}	11491288232933839952	c_{50}	−516953539510272
c_9	−9632413402	c_{23}	−365715901130565447	c_{37}	−16771570210364989920	c_{51}	−19841048084480
c_{10}	69606226129	c_{24}	944194941474727300	c_{38}	−5583493418068217728	c_{52}	14146663874560
c_{11}	111387705839	c_{25}	742089212374256992	c_{39}	9435286186908686336	c_{53}	2163980894208
c_{12}	−2254913344310	c_{26}	−3506287204334475748	c_{40}	2152938591505922240	c_{54}	93692362752
c_{13}	2790427012192	c_{27}	−244821434187388980	c_{41}	−3757256931413666560		
c_{14}	37993259179749	c_{28}	8829716447636956776	c_{42}	−719149914355861248		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 54, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $54 + 4$ terms removed returns order 54 as well, so $\deg q_{\min} = 54 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $54 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 54 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A237103.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.